Portfolio Optimization & Factor Analysis

1. Project overview

This project analyzes eight U.S. equities from January 2015 through December 2023. Monthly adjusted closing prices were converted to returns, and Excel Solver was used to construct a global minimum-variance portfolio, a maximum-Sharpe portfolio, and a sequence of frontier portfolios. The optimized portfolio was benchmarked against SPY and the S&P 500 and evaluated using market, Fama-French three-factor, and Carhart four-factor regressions.

2. Stock selection and descriptive statistics

The stock universe includes Cigna, Ford, IBM, Johnson & Johnson, Micron Technology, Sherwin-Williams, Warner Bros. Discovery, and Exxon Mobil. Monthly return distributions were summarized using mean return, standard deviation, skewness, excess kurtosis, and Jarque-Bera normality tests. The results show meaningful differences in volatility and distribution shape across the selected securities, with WBD and MU among the more volatile series and JNJ displaying comparatively lower volatility.

3. Portfolio optimization

A sample covariance matrix was estimated from monthly returns. Excel Solver minimized variance for the global minimum-variance portfolio and maximized the Sharpe ratio for the optimal risky portfolio, subject to weights summing to one and with short sales permitted. Eight additional portfolios were estimated at target risk-premium levels to map the minimum-variance frontier.

4. Optimized portfolio results

The global minimum-variance portfolio concentrated heavily in JNJ and achieved a monthly risk premium of approximately 0.771%, monthly volatility of 4.241%, and a monthly Sharpe ratio of 0.182. The maximum-Sharpe portfolio used positive allocations to CI, JNJ, MU, SHW, and XOM and short positions in Ford, IBM, and WBD. It achieved a monthly risk premium of approximately 1.587%, monthly volatility of 6.086%, and a monthly Sharpe ratio of 0.261.

5. Benchmark comparison

The maximum-Sharpe portfolio produced an annualized return of approximately 22.4% and annualized volatility of 22.8%, compared with 13.0% return and 13.0% volatility for SPY. The corresponding annualized Sharpe ratios were approximately 0.925 and 0.900. The optimized portfolio therefore generated the highest in-sample risk-adjusted performance, though its advantage over SPY was modest and depended on concentrated and short positions.

6. Market and factor regressions

The market regression estimated the relationship between optimized-portfolio excess returns and S&P 500 excess returns. The Fama-French model added market, size, and value factors, while the Carhart model added momentum. The multifactor specifications produced R-squared values of approximately 2%-3%, indicating that the selected factors explained little of the optimized portfolio's monthly return variation over the sample.

7. Individual-stock comparison

A Carhart four-factor regression was also estimated for Johnson & Johnson. Comparing the single-stock and portfolio models illustrates how diversification and portfolio weights can produce return patterns that differ from those of an individual constituent. Neither model displayed strong explanatory power from the included factors.

8. Interpretation and limitations

The results demonstrate the mechanics of covariance-based portfolio construction and factor analysis. The portfolio results are in-sample and do not include transaction costs, taxes, short-borrow fees, or out-of-sample validation. Mean-variance weights are sensitive to estimated returns and covariances, and the eight-stock universe is narrow relative to a realistic investment opportunity set.